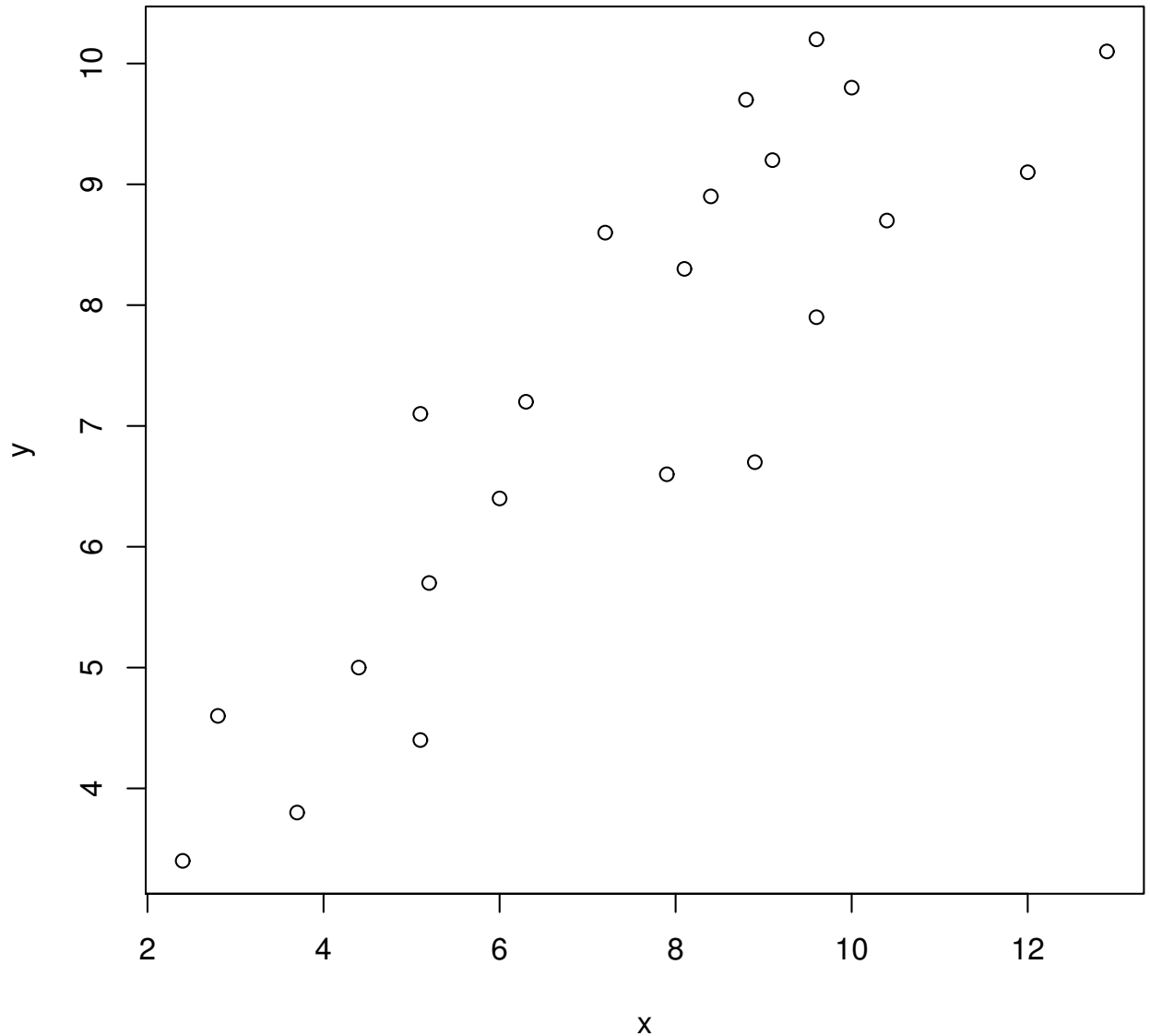
6-5. Multiple regression: Example 6.4

A model with a *qualitative* explanatory variable (factor).

- Data: $n = 22$ pairs (x_i, y_i) where y is the response.
- Data arise under two different conditions (type $= 1$ or 2).
- Presented below sorted by x within type.

Row	y	x	Type
1	3.4	2.4	1
2	4.6	2.8	1
3	3.8	3.7	1
4	5.0	4.4	1
⋮	⋮	⋮	⋮
19	9.7	8.8	2
20	9.2	9.1	2
21	10.2	9.6	2
22	9.8	10.0	2

```
> plot(y ~ x)
```



Example 6.4

First, fit model ignoring type.

```
> mod3 = lm(y ~ x); summary(mod3);
Call:
lm(formula = y ~ x)

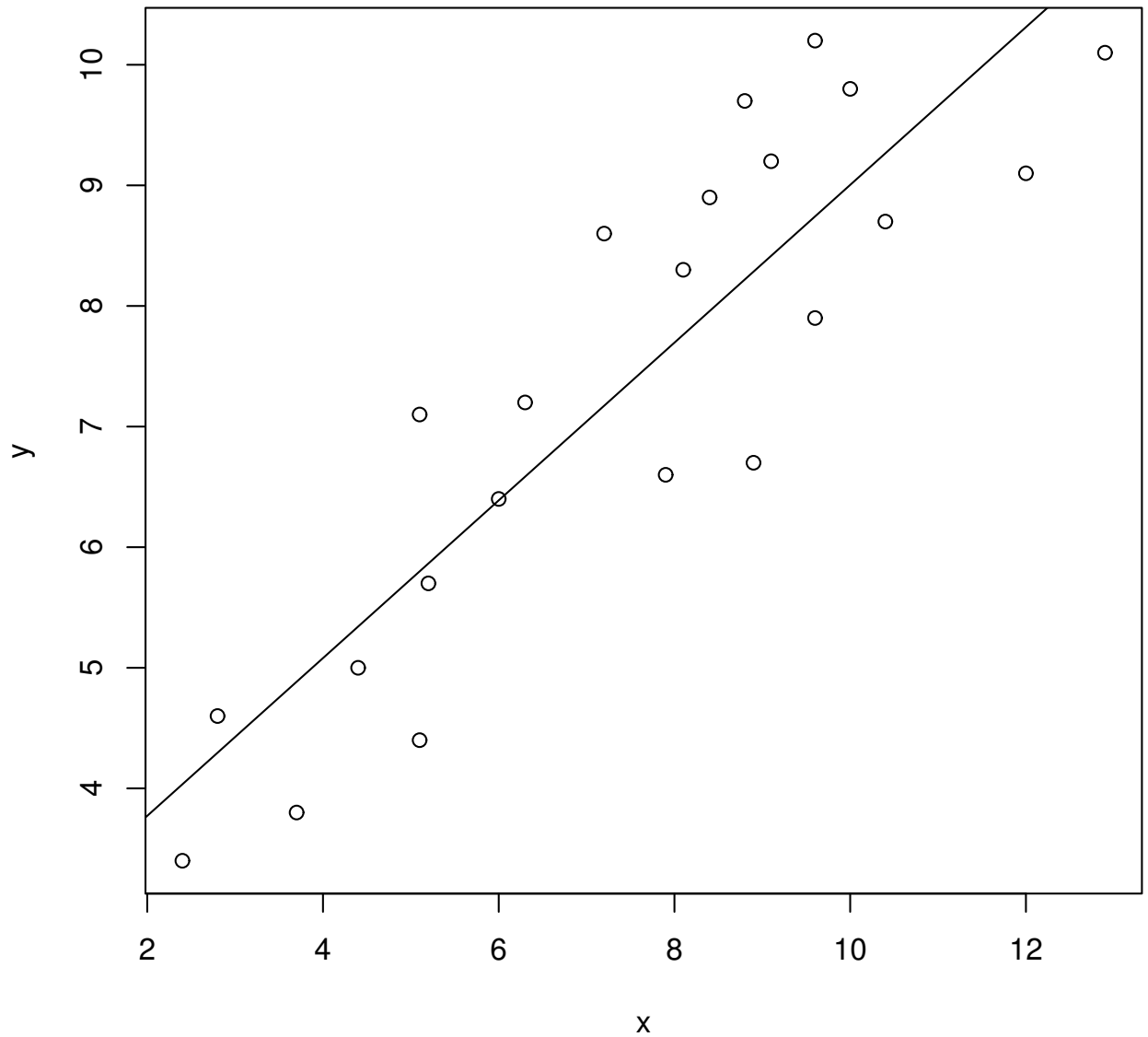
Residuals:
     Min       1Q   Median       3Q      Max
-1.58460 -0.83189 -0.07654  0.79318  1.48079

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.4644     0.6249   3.944 0.000803 ***
x             0.6540     0.0785   8.331  6.2e-08 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.033 on 20 degrees of freedom
Multiple R-squared:  0.7763,    Adjusted R-squared:  0.7651
F-statistic:  69.4 on 1 and 20 DF,  p-value: 6.201e-08
```

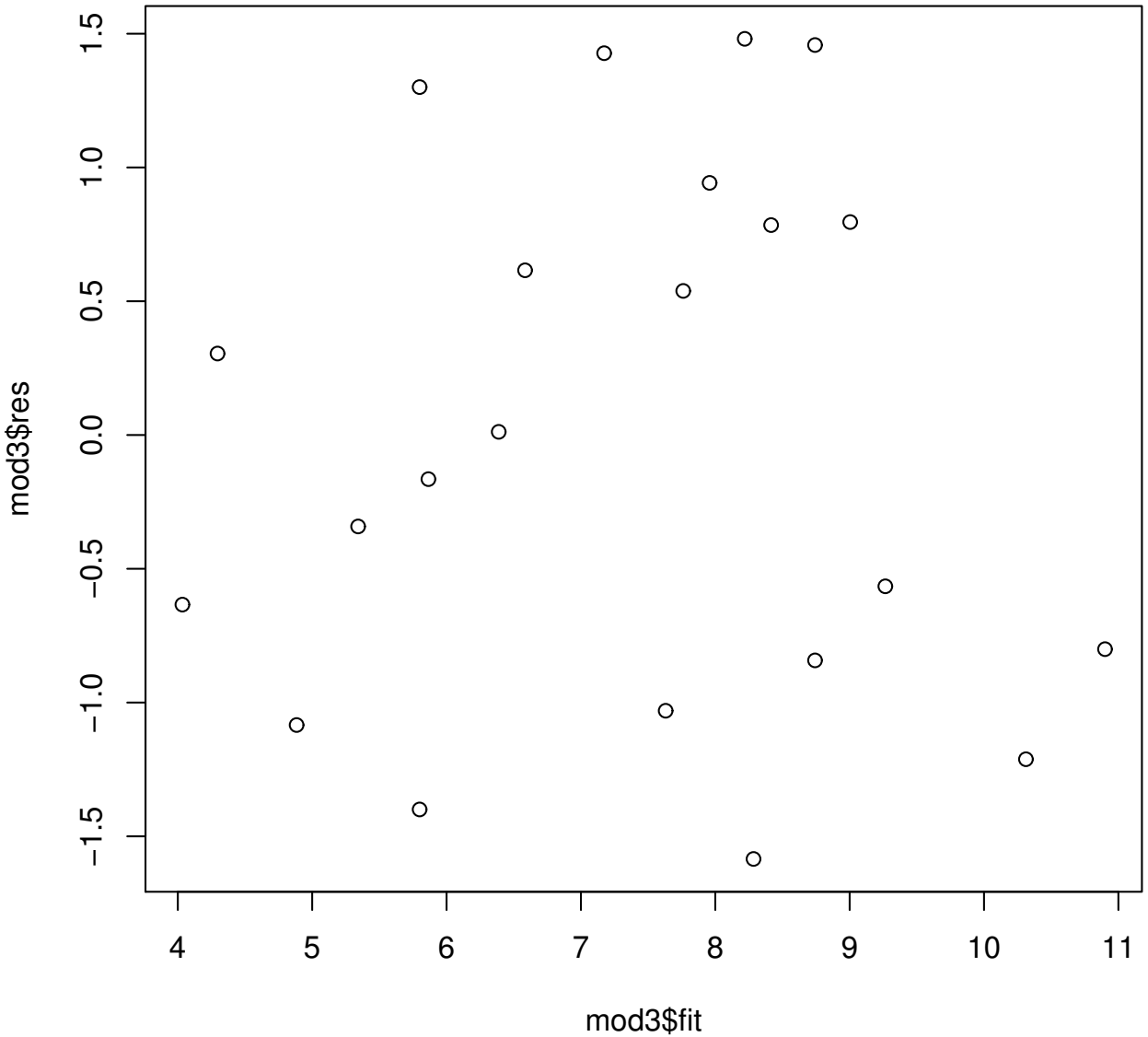
```
> summary.aov(mod3)
            Df Sum Sq Mean Sq F value  Pr(>F)
x            1  74.03   74.03    69.4 6.2e-08 ***
Residuals   20  21.34    1.07
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Plot data with fitted line:



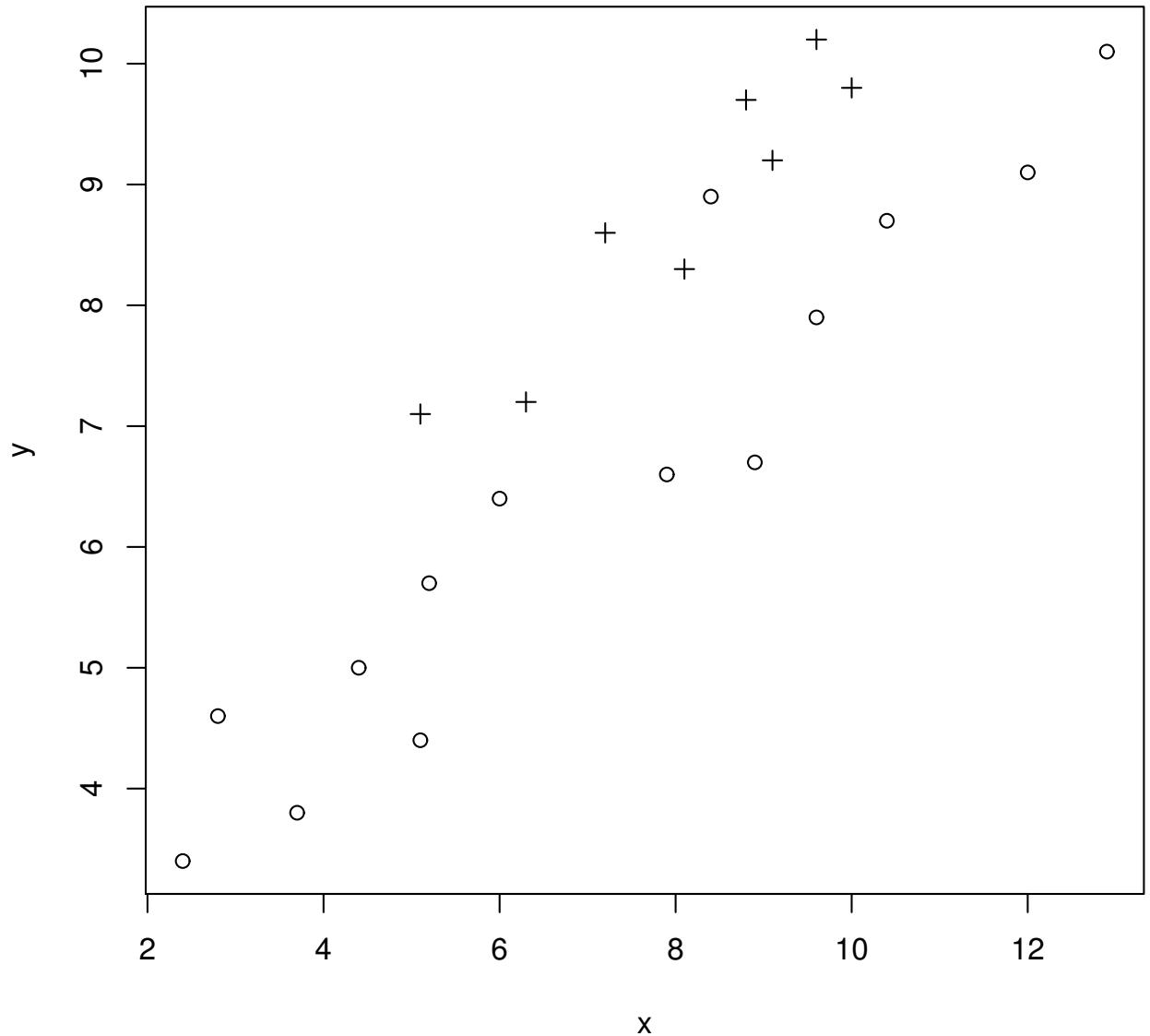
Example 6.4

Residuals versus fitted values:



Example 6.4

Distinguish the two types:



Now fit a model which includes the qualitative variable 'type'

To make sure that R treats the qualitative variable correctly, declare `type` as a factor:

```
> type = factor(type)
```

```
> mod4 = lm(y ~ x + type); summary(mod4);

Call:
lm(formula = y ~ x + type)

Residuals:
     Min       1Q   Median       3Q      Max
-0.90463 -0.39486 -0.03586  0.34657  1.59988

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.18426    0.37348   5.848 1.24e-05 ***
x            0.60903    0.04714  12.921 7.36e-11 ***
type2        1.69077    0.27486   6.151 6.52e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6127 on 19 degrees of freedom
Multiple R-squared:  0.9252,    Adjusted R-squared:  0.9173
F-statistic: 117.5 on 2 and 19 DF,  p-value: 2.001e-11
```

```
> summary.aov(mod4)

            Df Sum Sq Mean Sq F value   Pr(>F)
x            1  74.03   74.03  197.22 1.74e-11 ***
type         1  14.20   14.20   37.84 6.52e-06 ***
Residuals   19   7.13    0.38
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Fitted model is

$$\hat{y} \quad = \quad 2.18426 + 0.60903x \, (+1.69077 \text{ if type } 2)$$

eg Observation 1, with $x = 2.4$ and $type = 1$,

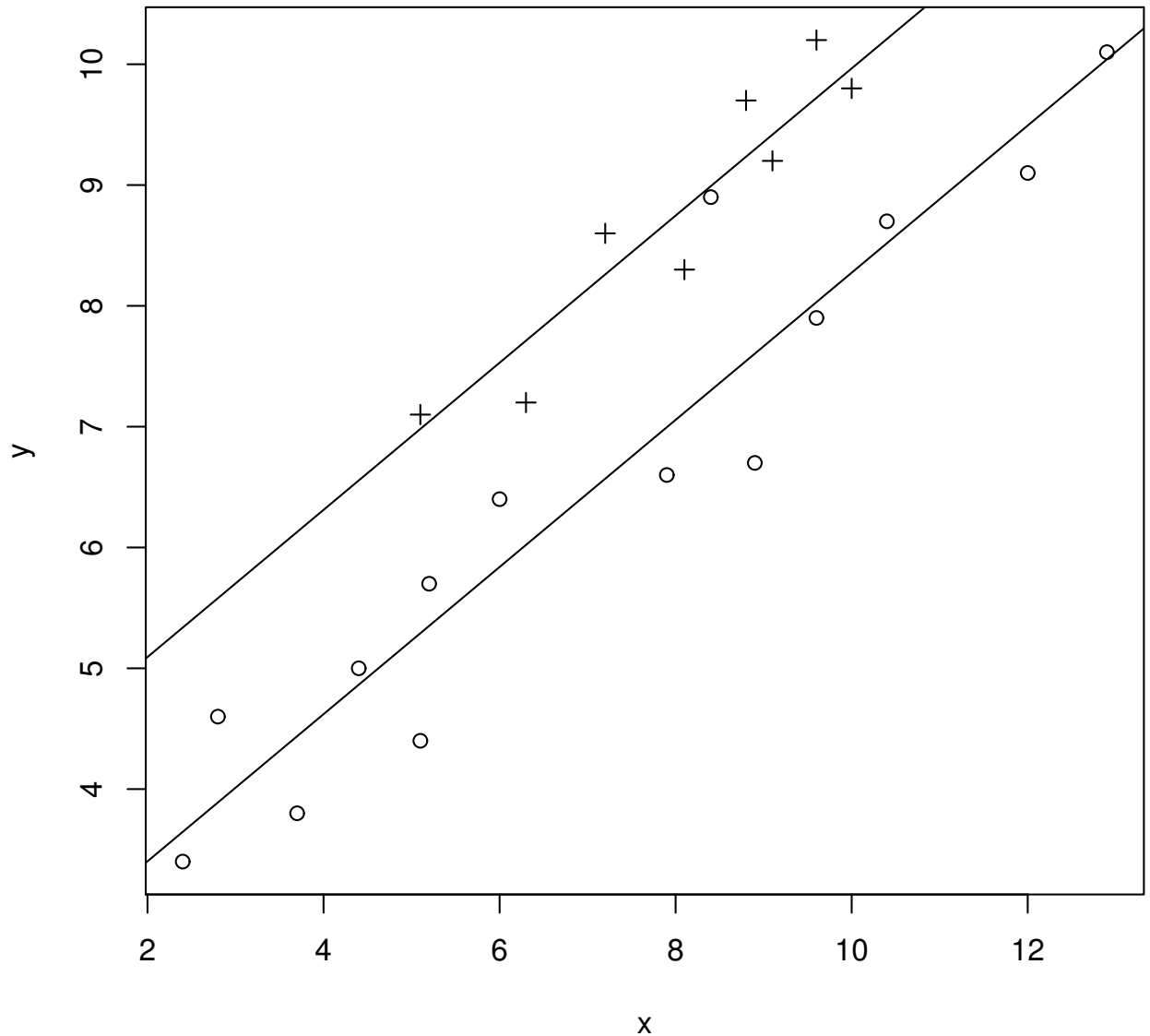
$$\hat{y}_1 \quad = \quad 2.18426 + 0.60903 \times 2.4 \quad = \quad 3.64593$$

eg Observation 20, with $x = 9.1$ and $type = 2$,

$$\hat{y}_{20} \quad = \quad 2.18426 + 0.60903 \times 9.1 + 1.69077 \quad = \quad 9.41720$$

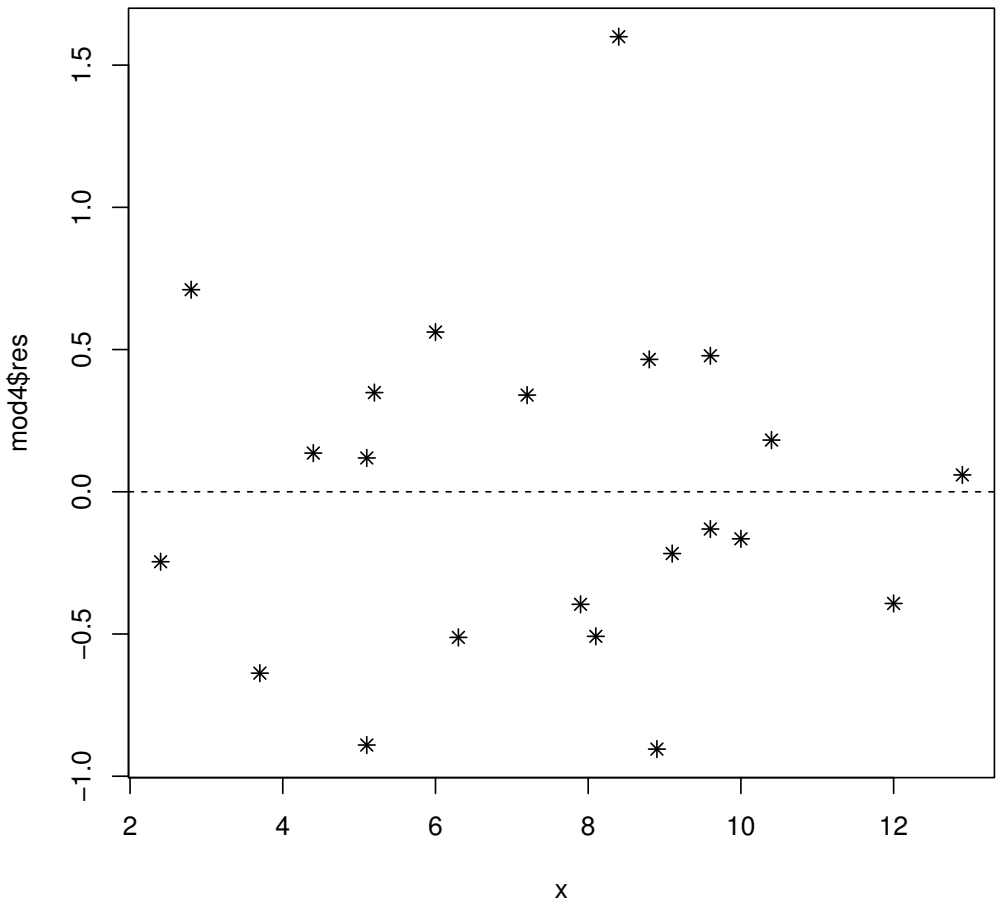
```
> fitted(mod4)
       1        2        3        4        5        6        7        8
3.645930 3.889543 4.437671 4.863993 5.290315 5.351218 5.838443 6.995603
       9       10       11       12       13       14       15       16
7.300119 7.604634 8.030956 8.518181 9.492632 10.040760 6.981083 7.711921
      17       18       19       20       21       22
8.260049 8.808177 9.234499 9.417209 9.721724 9.965337
```

Fitted lines for type 1 and type 2 have same slope, different intercepts.



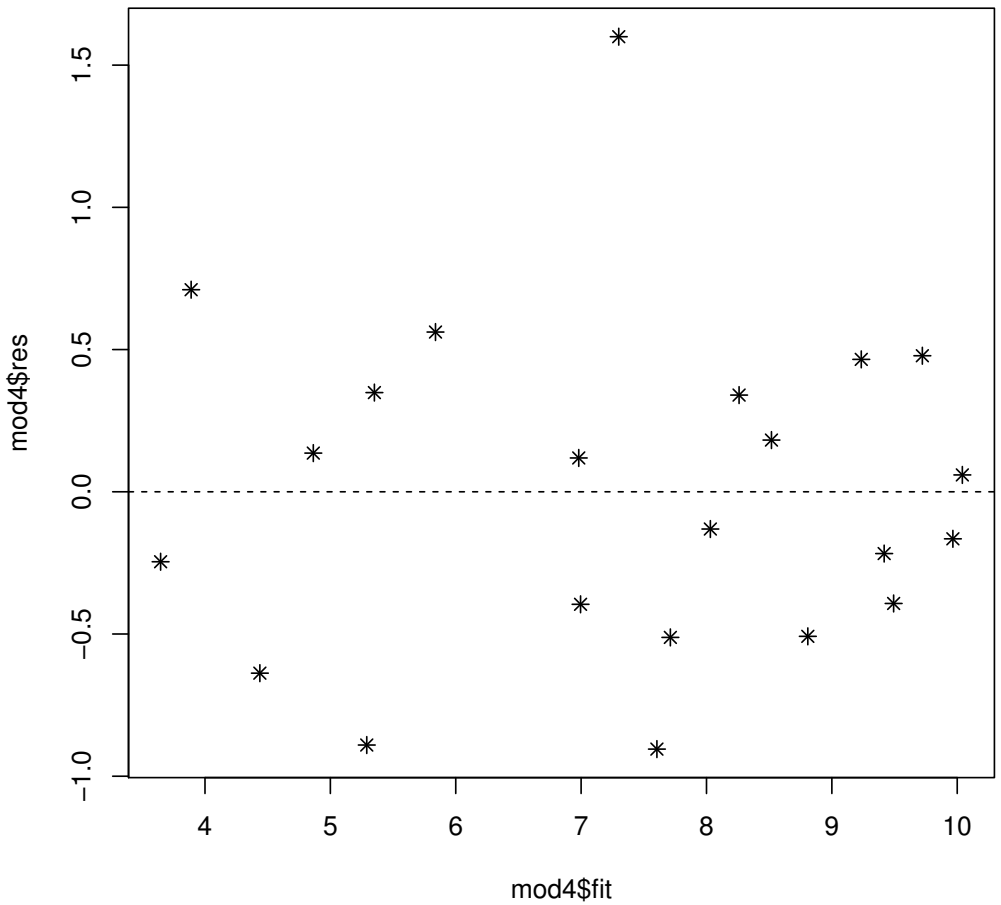
Plot residuals versus x values.

```
> plot(x, mod4$res, pch=8)
> abline(0,0,lty="dashed")
```



Plot residuals versus fitted values.

```
> plot(mod4$fit, mod4$res, pch=8)
> abline(0,0,lty="dashed")
```



Interaction

- Finally, we can extend the model to allow different slopes (as well as different intercepts) for the two types.
- That is, allow an *interaction* between the explanatory variables x and *type*.
- An interaction exists between two explanatory variables when **the effect of one on the response variable is different at different values/levels of the other**.
- For example consider the effect of policyholder's *age* and *gender* on a response variable *claim rate.*
  If the effect of age on *claim rate* is different for males and females, then there is an interaction between *age* and *gender*.

Using data of example 6.4, fit model with interaction between x and $type$.

```
> mod5 = lm(y ~ x * type); summary(mod5);

Call:
lm(formula = y ~ x * type)

Residuals:
     Min       1Q   Median       3Q      Max
-0.90080 -0.38551 -0.01445  0.36309  1.60651

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.22119    0.40345   5.506 3.15e-05 ***
x            0.60385    0.05152  11.721 7.36e-10 ***
type2        1.35000    1.20826   1.117    0.279
x:type2      0.04305    0.14843   0.290    0.775
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.628 on 18 degrees of freedom
Multiple R-squared:  0.9256,    Adjusted R-squared:  0.9132
F-statistic:  74.6 on 3 and 18 DF,  p-value: 2.388e-10
```

```
> summary.aov(mod5)
            Df Sum Sq Mean Sq F value   Pr(>F)
x            1  74.03   74.03 187.715 5.81e-11 ***
type         1  14.20   14.20  36.014 1.12e-05 ***
x:type       1   0.03    0.03   0.084    0.775
Residuals   18   7.10    0.39
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

- In this particular case, an interaction term is not helpful.
- In a case where an interaction term does improve the fit and the coefficient is significant, then both variables and the interaction between them should be included in the model.